EARTH'S FIELD NMR -- a proton precession spectrometer

A coil round a bottle of water. POLARIZE: a few amperes through the coil for a few seconds magnetize the water's protons along the coil's field, a few hundred times more than the Earth's 46 uT alone would.
Switch off: in a millisecond the field is gone, faster than the protons can follow, and they precess about the Earth's field at f = 42.58 Hz/uT x B: 1960 Hz here, a few microvolts in the same coil, dying away in a fraction of a second to a few seconds (T2*). LISTEN: the relay hands the coil to a tank tuned to f, and the preamplifier and the filters raise the microvolts to a volt.
The reference coil (handbook): 916 turns of AWG 20 on 3-inch pipe, 27 mH, 8.9 ohm, Q 37. At 1.5 A: 7.8 mT, about 1.2 uV in the coil from 1 L of water, 45 uV across the tank, 0.9 V at OUT on HIGH gain.

POLARIZING SUPPLY

6-30 V, up to 3 A. A bench supply in constant current sets the coil's current; or the coil's resistance does: the reference coil draws 3 A at 26 V.

V POL: behind the 3 A fuse

LED3: the supply is there

COIL

A (2), B (1): the coil's two leads
S (3): its shield, if any
up to 130 V at switch-off:
do not touch
10-200 mH, up to 3 A, 1/2 L I^2 < 1 J

K1 HFD3/12: released, the coil LISTENS (as drawn); energized, it is POLARIZED: A to V POL, B to the switch

DRAIN: up to 160 V, a x10 probe

I POL: 0.1 V per ampere, on and ramping off

D7 SMCJ100A holds the drain near 130 V while the coil's current falls to zero in a straight line (0.4 ms for the reference coil): faster than the protons can follow

D9: a clamp, idle unless a fault dumps the coil's energy here

CAL IN

a generator through 1M: I = V / 1M into the tank. mV-level, 5 Vpp max

SW1 TUNE: each switch adds its capacitance (C0G) to the tank, in 3.3 nF steps up to 841 nF.
Resonance: f = 1 / (2 pi sqrt(L C)); tune to the Larmor frequency for the most signal.
The reference coil needs 246 nF at 1960 Hz. Caps C10-C29: 3.3n, 6.8n, 10n 0805, 47n, 100n 1206

ground loops for the scope's clip

PREAMPLIFIER

an instrumentation amplifier, gain 101: U1 and U2 amplify each side of the tank, U3 takes the difference, so pickup common to both cancels.
OPA197: 5.5 nV/rtHz and no current noise against the tank's 12 k: 1.7 dB noise figure with the reference coil.
R11, R12 (1M) give the inputs their DC; R15, R16 (1k) limit a fault's current into them

gain 101 = 1 + 2 x 10k / 200R
(between the two - inputs)

HIGH-PASS

4th-order Butterworth, 688 Hz: two unity-gain Sallen-Key sections, Q 0.55 at 726 Hz and Q 1.31 at 670 Hz. 60 Hz -85 dB, 180 Hz -47 dB, 420 Hz -17 dB: mains hum and its low harmonics go. C0G capacitors.

LOW-PASS

2nd-order Butterworth, 4097 Hz (Q 0.72): the top of the band, and the anti-alias filter for a sound card. C0G capacitors.

GAIN

x21 or x201 = 1 + 200k / Rg

SW2 GAIN
1: LOW, x2121 overall (67 dB)
3: HIGH, x20301 (86 dB)

OUT

the FID, about 1 V on HIGH;
1k in series: for a scope, an ADC or a sound card's line input

HEADPHONES

a mono copy of OUT for a stereo headset: 30 mV of FID at a 32 ohm earpiece on HIGH, the knob full up (about 81 dB SPL); R56, R57 hold a mono plug's shorted ring to 21 mA. J9: Same Sky SJ1-3523N, 1 sleeve, 2 tip, 3 ring

VOLUME
clockwise louder

POWER

+15 V, GND, -15 V,
100 mA a rail

D1, D2: a reversed supply does nothing
D3, D4: clamp a rail left open
LED1, LED2: both rails present

SEQUENCER

POL CMD high = POLARIZE: the relay pulls in at once, and the switch closes 18 ms later (U9).
POL CMD low: the switch opens at once, the coil's current ramps off through D7, and the relay lets go 15 ms later (U10): LISTEN.

RV1 POLARIZE TIME
0.2-4.7 s, clockwise longer

RV2 LISTEN TIME
0.2-4.8 s, clockwise longer

C50-C52: 3 x 1 uF polypropylene film; t = 0.75 R C each way

SW3 SEQUENCE
1: EXT, the POLARIZE IN jack
3: INT, the timer (RV1, RV2)

SW4 MODE
1: RUN
3: LISTEN ONLY (no polarizing)

U9 GATE: on 18 ms after POL CMD rises (R49, C55), off at once when it falls (D13)

U10 RELAY: on at once (D14), off 15 ms after POL CMD falls (R50, C56)

K1 coil
1028 ohm

red: POLARIZING

POLARIZE IN

high (> 1.3 V) = polarize: TTL, 3.3 V, 5 V, a function generator; ±30 V max; open = listen

U8: a comparator at 1.3 V (R45, R46), 147 mV of hysteresis (R47).
D12 clamps the input between -0.35 V and +15.3 V; R59 keeps the TS391's pin inside its -0.3 V limit at ±30 V in

SYNC OUT

follows POL CMD: 0 / 13 V open, through 1k; its falling edge is the switch-off: trigger the scope on it

M3 mounting holes, one per corner

Bypass: 100 nF X7R at every op-amp's, comparator's and the timer's supply pins (C80-C99)

Open source, CERN-OHL-P-2.0 -- github.com/gallicchio/earth-nmr-pcb
OPA197 for every amplifier, TS391 comparators, NE555 timer, IRFS4227 switch, HFD3 relay
937-2853 Hz anywhere on Earth
Larmor frequency 1960 Hz at Harvey Mudd College, Claremont CA (46.0 uT, IGRF-14, 2026-11-01)
Jason Gallicchio and Claude
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